Ines Vati

InesVati | i.vati@qut.edu.au | ines-vati | +61 4 86 03 12 74
inesvati.github.io | Brisbane, Australia

Seeking a research internship to solve complex problems in Machine Learning with application to vision and/or language processing, and large-scale AI systems

EDUCATION

PhD in Computer Vision and Machine Learning (CU) — Feb. 2025 – Present
Queensland University of Technology & CSIRO — Brisbane, QLD

- **Thesis Title**: *Fast Geometric Machine Learning Methods for Brain Image Analysis*
- **Research**: Project being conducted within both the Health and Biosecurity Unit at CSIRO and the Signal Processing, Artificial Intelligence and Vision Technologies (SAIVT) Lab at QUT
- **Objective**: Enhancing the accuracy and efficiency of cortical surface analysis
- **Keywords**: Graph Neural Networks, Implicit Neural Representation, Surface Registration

MVA Research Master (Mathematics, Vision, Learning) — Oct. 2023 – Oct. 2024
ENS Paris-Saclay — Paris, FR

- **Description**: Masters degree in Computer Vision, Data Science and Machine Learning
- **Coursework**: Computational Statistics, Convex Optimization, Computational Optimal Transport, Kernel Methods, Point Cloud and 3D Modelisation

Engineering Degree in Mathematics & Computer Science — Sept. 2020 – Oct. 2024
École des Ponts ParisTech — Champs-sur-Marne, FR

- **Coursework**: Random processes, Software development techniques, Deep Learning, Statistics and Data Analysis, Image processing and Artificial Vision

EXPERIENCE

Machine Learning Postgraduate Internship — Apr. 2024 – Oct. 2024
CSIRO — Brisbane, QLD

- Conducted an **in-depth literature review** of the state-of-the-art methods in brain surface analysis and parcellation
- Developed innovative approaches using **Geometric Deep Learning** methods and **Spectral Graph theory** to enhance brain surface processing and their parcellation

Medical Imaging Research Internship in Industry — Feb. 2023 – July 2023
Dassault Systèmes — Vélizy-Villacoublay, FR

- Implemented and experimented with convolution-based neural networks, such as VoxelMorph, for **3D medical image registration** aiming at reconstructing 3D anatomical structures of organs
- Developed innovative models for **learning anatomical templates**, capturing the variability within a given patient population
- Participated in an internal **poster session**, showcasing research findings and engaging with peers

Modeling and Deep Learning Internship — Sept. 2022 – Jan. 2023
MaIAGE INRAE — Jouy-en-josas, FR

- Developed mathematical models of dengue epidemic dynamics; Used for simulation and generation of phylogenetic trees
- Implemented and trained neural networks to infer epidemiological parameters of complex models; Achieved competitive results compared to the state-of-the-art
- Presented research outcomes during the lab' s weekly **seminar**

TEACHING POSITIONS

Teaching Assitant in CAB420 Machine Learning Feb. 2026 – Present
Queensland University of Technology Brisbane, QLD

- *Type*: Undergraduate course from the Faculty of Engineering at QUT
- Introduced machine learning methods such as linear regression, dimension reduction techniques, metric learning, etc.
- Solved classical problems such as image classification, semantic segmentation, multi-task learning, retrieval and ranking, etc.
- Presented tutorials using JupyterLab, Python and librairies such as `statsmodels`, `scikit-learn`, and Keras

Teaching Assitant in QUT001 Artificial Intelligence in the Real World Feb. 2025 – Apr. 2025
Queensland University of Technology Brisbane, QLD

- *Type*: Undergraduate course from the Faculty of Engineering at QUT
- Explored with students from different faculties (Business, Health, Law, etc.) what Artificial Intelligent is, how it is developed, and its potential pitfalls
- Raised awareness about algorithmic bias, Ethical AI, and AI' s impact on their future career

PUBLICATIONS

Conference Paper

- Vati, Ines, Pierrick Bourgeat, Rodrigo Santa Cruz, Vincent Dore, Olivier Salvado, Clinton Fookes, and Léo Lebrat. "NC-Reg: Neural Cortical Maps for Rigid Registration." arXiv e-prints (2026): arXiv-2601. (Accepted at the *2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*)

Abstract

- Javier Antonio Urriola Yaksic, Leo Lebrat, Hang Min, Alex Pagnozzi, Ying Xia, Ines Vati, Charles X. Li, Senali Weeratunga, Merran Courney, Jurgen Fripp, Matthew Gutman, Andrew Neal, Ben Sinclair, Terrence J. O'Brien and DanaKai Bradford, "Advanced Deep Learning Models Enhance Piriform Cortex Segmentation for Epilepsy Surgery", 2025 *ISMRM & ISMRT*

Talks & Presentations

Oral Presentation at International Conferences

- "NC-Reg: Neural Cortical Maps for Rigid Registration", Oral presentation, IEEE 23rd International Symposium on Biomedical Imaging (ISBI), 2026, London, UK

Invited Presentations & Seminars

- Invited Talk, HeKA Inria team, 2026, PariSanté Campus, FR
- Invited talk, NeuroSpin, 2026, CEA Paris-Saclay, FR
- Seminar presentation, Scientific seminar series DARE U (Dassault Systèmes Research Universe) at Dassault Systèmes, 2026, Vélizy-Villacoublay, FR
- "Fast Geometric Deep Learning to streamline cortical surface processing", Applied Mathematics Seminar, CERMICS lab, 2026, Champs-sur-Marne, FR

PERSONAL PROJECTS

EPVS Segmentation Challenge - MICCAI2024 2024

- *Problem*: Segmentation of Enlarged Perivascular Spaces (EPVS) in brain MRIs of different modalities and from multiple sites and scanners
- Collaborated with talented researchers from my lab during my postgraduate internship
- Achieved **1st place**

Ocean Hackathon 6 2021

- 48 hours to develop a prototype application for raising awareness about marine biodiversity using various sea-related data
- Developed a prototype for an app that enables users to identity any fish species through photo recognition
- Pitched our product to a jury and Achieved 1st place for the pre-selection phase at ENSG in Champs-sur-Marne

ACADEMIC PROJECTS

Kernel Methods Kaggle Challenge 2024

- Participated in the MVA course "*Machine Learning with Kernel Methods*" data challenge
- Focused on multi-class image classification on pre-processed images
- Achieved 2nd place on the leaderboard

Medical Image Analysis Project 2023

- Tackled highly unbalanced segmentation in medical images
- Conducted a study on the boundary-based loss function proposed by Kervadec et al.
- Written a technical report and delivered a concise presentation

MOPSI (MOdel Program SImulate) Project 2022

- Collaborated in a 4-months project within a team of three
- Explored machine learning and dimension reduction algorithms for **molecular simulation**
- Developed an interactive website explaining key concepts, and allowing users to visualize and simulate molecular trajectories
- Presented our work in a poster session

SKILLS

Programming Languages : Python, R, C++, Julia
Tools : Linux, Git, MATLAB, HPC (GPU clusters, SLURM), FreeSurfer, Docker, HuggingFace
Communication : English (C1), French (Native), Spanish (B2), Arabic (A2)